Institution's Innovation Council
MHRD's Innovation Cell, AICTE

Innovation / Prototype
(Product/Service/Process)

Team Details

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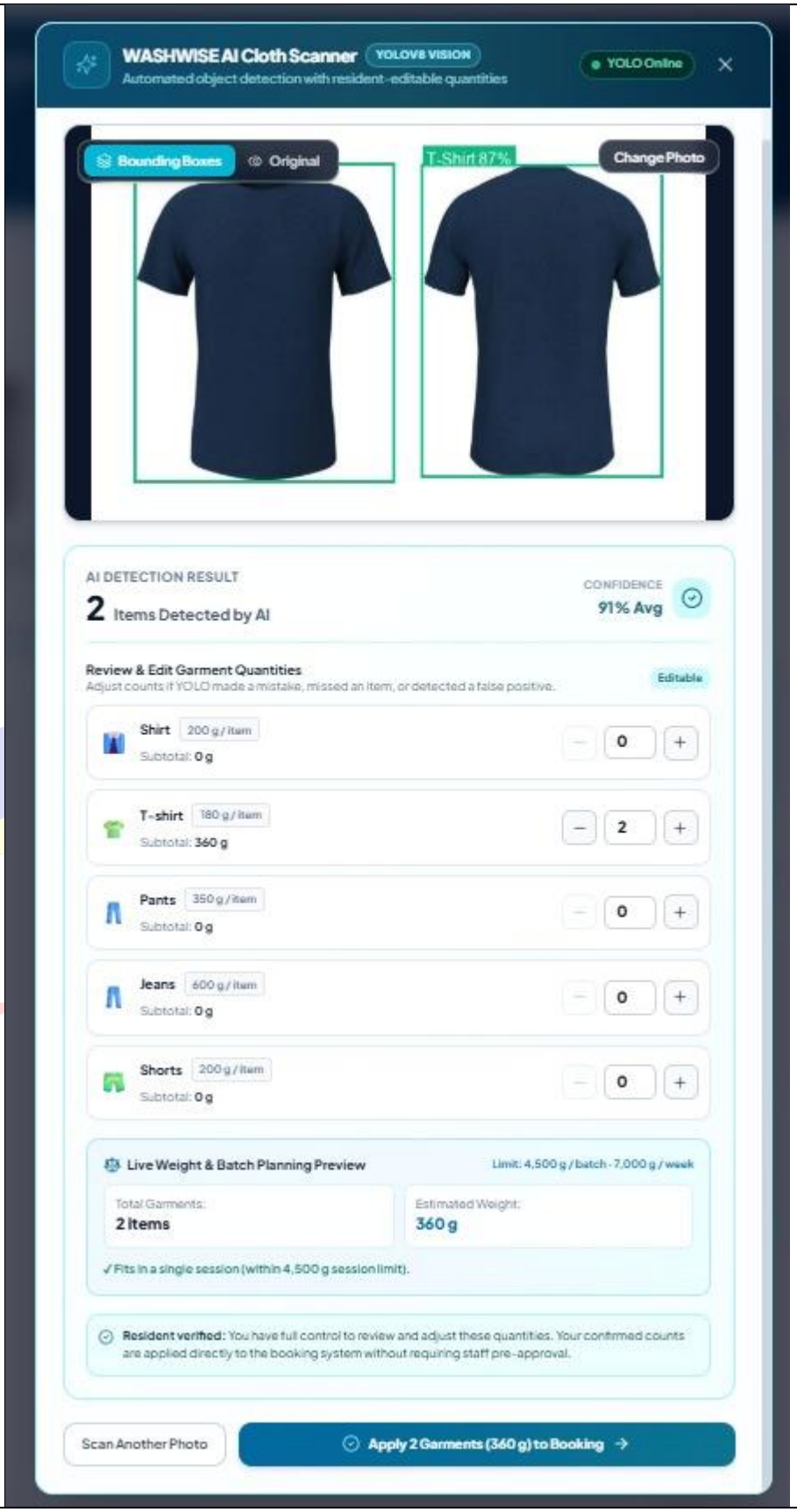
Title (20 Words Max)	WASHWISE – SMART HOSTEL LAUNDRY MANAGEMENT WITH YOLO-BASED CLOTHING DETECTION
***Developed as part of**	Academic Requirement / Study Project
***Choose the Financial Year, during the Idea-PoC/Innovation Developed**	2026-2027
***Sector / Domain**	Information and Communication Technology (ICT) - Artificial Intelligence (AI) & Machine Learning (ML)
Innovation Type	Product
***Development Stage - Technology Maturity of the Solution/Innovation in terms of Technology Readiness Level TRL (if applicable (Refer TRL Stages)**	TRL 6 - Prototype system tested in intended environment close to expected performance

Development Stage - Manufacturing Maturity of the Solution/Innovation in terms of Manufacturing Readiness Level (MRL) Refer MRL Stages	Not Applicable
Development Stage: Investment Readiness Level of the Solution/Innovation (IRL). Refer IRL Stages	IRL 4 – Prototype Low-Fidelity Minimum Viable Product (MVP)
***Define the problem and its relevance to today's market / society / industry need (Max: 100 Words)**	Many college hostels and PG accommodations still depend on manual laundry management, which can lead to long waiting times, inefficient machine usage, poor tracking of laundry activities and difficulty managing multiple users. Existing manual methods also lack accurate garment counting and proper booking management. With the increasing demand for convenient and efficient laundry services among students and hostel residents, there is a need for a smart, automated solution. WASHWISE addresses these challenges through digital laundry booking, machine and slot management, and AI-based clothing detection using YOLO technology.
***Describe the Solution / Proposed / Developed (Max: 100 Words)**	WASHWISE is a smart laundry management platform designed for college hostels and PG accommodations. It provides separate login facilities for college and PG hostel users, allowing residents to book laundry slots, manage clothing quantities and track their laundry activities. The system uses YOLO-based artificial intelligence to detect different clothing types and count garments through a camera. It also provides automated slot allocation, washing machine availability management and weekly laundry usage tracking. The platform aims to make laundry operations more convenient, organized and efficient through a user-friendly digital system.
***Explain the uniqueness and distinctive features of the (product / process / service) solution (Max: 100 Words)**	The main uniqueness of WASHWISE is its integration of YOLO-based AI clothing detection into a smart laundry management system. It can identify different clothing types and automatically count garments using a camera, reducing the need for manual counting. The platform also provides separate access for college and PG hostel users, intelligent laundry slot allocation, washing machine availability management and weekly usage limits. By combining computer vision with digital booking and laundry tracking, WASHWISE offers a more organized, convenient and technology-driven approach to hostel laundry management.

***How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any (Max: 100 Words)**	Unlike conventional laundry management systems that mainly focus on booking and payment facilities, WASHWISE integrates YOLO-based AI technology to detect clothing types and automatically count garments using a camera. It is specifically designed to address the laundry management needs of both college hostels and PG accommodations through separate user access. Its additional features include intelligent slot allocation, washing machine availability tracking, session and weekly usage limits, and laundry activity monitoring. This combination of AI-based garment detection and hostel-focused laundry management makes WASHWISE a distinctive solution.
Is there any IP or Patentable Component associated with the Solution?	Currently, WASHWISE does not have any registered patent or intellectual property rights. However, its software implementation, system design and integration of YOLO-based AI clothing detection represent original project development that may be considered for future intellectual property protection.
Has the Solution Received any Innovation Grant/Seed fund Support?	WASHWISE has not received any innovation grant, seed funding or external financial support so far. The project has been developed as an academic initiative using the available institutional and team resources. Its development and implementation have been carried out without external funding assistance.
Are there any Recognitions (National/International) Obtained by the Solution?	WASHWISE has not yet received any national or international awards, recognitions or formal certifications. However, the project has been developed as an academic innovation, demonstrating the practical application of artificial intelligence and modern web technologies to address real-world laundry management challenges.
Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?	WASHWISE has not yet been commercialized through technology transfer or established as a startup or enterprise. It is currently developed as an academic project and functional prototype. However, the solution has potential for future commercialization through partnerships with colleges, hostels and PG accommodation providers.
Had the Solution Received any Pre-Incubation/Incubation Support?	WASHWISE has not received any formal pre-incubation or incubation support from any startup incubator or innovation centre so far. The project has been developed independently as an academic initiative with the efforts of the project team and available institutional resources.
***Utility: Highlight the utility/value proposition (key benefits) aspects of the solution/innovation* (Max: 100 Words)**	WASHWISE provides a convenient and efficient way to manage laundry activities in college hostels and PG accommodations. Its AI-based YOLO clothing detection reduces manual effort by identifying clothing types and counting garments automatically. Digital laundry booking helps users select available washing slots and avoid

	unnecessary waiting. The system also manages machine availability, session limits and weekly laundry usage, ensuring organized resource utilization. Separate access for college and PG users makes the platform suitable for different accommodation environments, improving convenience, time management and the overall laundry experience.
***Scalability: Highlight the market potential aspects of the Solution/Innovation (Potential Market Size, segmentation and Target users/customers etc.) (Max: 100 Words)**	WASHWISE has strong market potential among educational institutions, college hostels, private hostels, PG accommodations and student residential facilities. Its primary target users are students, hostel residents and accommodation administrators who require an organized laundry management system. The platform can be expanded to multiple colleges and PG facilities by adapting its user management and laundry operations to different environments. With increasing demand for digital solutions and automated services, WASHWISE has the potential to serve a wider market across India and can be further developed for large-scale adoption in residential communities and institutional facilities.
***Economic Sustainability: Highlight commercialisation/business application aspects of the solution (how it is going to economic profitable and viable) (Max: 100 Words)**	WASHWISE has the potential to become an economically viable solution through subscription-based services offered to colleges, hostels and PG accommodation providers. Institutions can adopt the platform to streamline laundry operations, reduce manual management efforts and improve the efficient use of washing machines. Revenue can be generated through institutional licensing, annual subscriptions and customized software services. As a software-based solution, it requires relatively low additional infrastructure for expansion, making it suitable for future commercialization. Its adaptable design also provides opportunities for long-term growth and adoption across multiple institutions.
***Environmental Sustainability: Highlight environmental friendliness aspects and related benefit of the solution/innovation (Max: 100 Words)**	WASHWISE promotes environmental sustainability by encouraging efficient utilization of washing machines and better management of laundry activities. Its intelligent slot allocation and laundry load management help reduce unnecessary machine usage and avoid inefficient washing cycles. By organizing laundry operations and monitoring clothing quantities, the system supports more responsible use of water and electricity. Its digital platform also reduces dependence on paper-based booking and manual record maintenance. Overall, WASHWISE contributes to sustainable resource management by promoting organized laundry practices and supporting environmentally conscious hostel operations.
Video URL	

Upload Photograph: (JPG, PNG, PDF max 2 MB)



(e) Timeline- Develop a timeline against the milestones for taking Idea/PoC to Prototype Development and (or) Commercial level/start-up stage.	**Phase 1 – Idea and Planning (July 2026):** Identified laundry management challenges in college hostels and PG accommodations and finalized the project concept. **Phase 2 – Design and Development (August 2026):** Designed the user interface, database and backend, and developed the core laundry booking and management features. **Phase 3 – AI Integration (September 2026):** Integrated YOLO-based clothing detection to identify clothing types and automatically count garments. **Phase 4 – Testing and Deployment (October 2026):** Tested the complete system, verified its major functionalities and deployed the application using cloud services. **Phase 5 – Future Commercialization:** Plan to expand WASHWISE to more colleges, hostels and PG accommodations through institutional partnerships and subscription-based services.
Applicability of Solution 10 Marks *(Max 50 words for each from a-e)*	
(a) **Usability:** what is the usability of your innovation. *(Level of acceptance of innovation and its Features among target group)*	WASHWISE offers a user-friendly platform that simplifies laundry booking and management for students and hostel residents. Its AI-based clothing detection, easy slot booking and laundry tracking features reduce manual effort and save time. The simple interface makes it convenient for students, administrators and hostel management to use.
(b) **Scalability:** how your innovation will be scalable at market level.	WASHWISE can be expanded to multiple colleges, private hostels and PG accommodations by adapting its user management and laundry facilities. Its modular software architecture allows deployment across different institutions, making it suitable for wider adoption in the educational and residential accommodation sectors.
(c) **Economic sustainability:** Explain the potential of innovation to become profitable or financially viable.	WASHWISE has the potential to generate revenue through subscription-based services and institutional licensing for colleges, hostels and PG accommodations. Its software-based model requires relatively low maintenance and expansion costs, making it a financially viable solution with opportunities for long-term growth and commercialization.
(d) **Environment Sustainability:** How your innovation is environment friendly or address environmental problems.	WASHWISE promotes environmental sustainability by encouraging efficient use of washing machines and better laundry load management. Its digital booking system reduces paper usage, while organized slot allocation helps avoid unnecessary machine operations. This supports responsible utilization of water and electricity and encourages sustainable laundry practices in hostels.

(e) Is there any Intellectual Property (IP) Component associated with innovation? if yes, explain.	Currently, WASHWISE does not have any registered intellectual property or patented components. However, its software design, system architecture and integration of YOLO-based AI clothing detection represent original development work that may be considered for future intellectual property protection.
Define the potential market size (in terms of INR) and target customers. *(Max 100 words)*	WASHWISE initially targets a potential market of approximately ₹5 lakh through adoption by small and medium-sized colleges, private hostels and PG accommodations. Its primary customers include educational institutions, hostel administrators and PG operators. Students and hostel residents will be the main users. The platform can gradually expand to more institutions through affordable subscription plans and institutional partnerships, creating opportunities for future growth in the Indian market.

INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

Annexure 1

Themes:

1. Healthcare & Biomedical devices.
2. Agriculture & Rural Development.
3. Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology.
4. Food Processing.
5. Robotics and Drones.
6. Waste management.
7. Clean & Potable water.
8. Renewable and affordable Energy.
9. IoT based technologies (e.g. Security & Surveillance systems etc)
10. ICT, cyber physical systems, Block chain, Cognitive computing, Cloud computing, AI & ML.

9 stages of TRL:

TRL 0 : Idea. Unproven concept, no testing has been performed.

TRL 1 : Basic research. Principles postulated observed but no experimental proof available. TRL 2 : Technology formulation. Concept and application have been formulated.

TRL 3 : Applied research. First laboratory tests completed; proof of concept.

TRL 4 : Small scale prototype built in a laboratory environment ("ugly" prototype). TRL 5 : Large scale prototype tested in intended environment.

TRL 6 : Prototype system tested in intended environment close to expected performance. TRL 7 : Demonstration system operating in operational environment at pre-commercial scale. TRL 8 : First of a kind commercial system. Manufacturing issues solved.

TRL 9 : Full commercial application, technology available for consumers.